COFFEE SHOP NETWORK

Secure Small-Business Network Design and Configuration

VLANs | Trunking | Router-on-a-Stick | DHCP | Guest ACL | SSH

Note: addressing values (subnets, some port numbers, DNS, SSID) are a documented reference plan. Verify them against your Packet Tracer file.

01 - Requirements and Design

Client Brief

Mike is opening a neighborhood coffee shop next week with no network in place. He needs:

A secure office network for running the business (payroll, invoices, schedules, reports).

A reliable POS system to process customer payments.

Free Wi-Fi for customers that cannot reach business resources.

Site Survey

Area	Equipment	Business Need	Network Decision
Manager's office	Manager PC, office printer	Confidential data (payroll, invoices, schedules)	Management Office VLAN
Checkout counter	POS terminal, receipt printer	Payment processing, the most critical system	Dedicated POS VLAN
Seating area	Wireless access point	Customer internet access	Isolated Guest Wi-Fi VLAN
Network cabinet	Router, switch, ISP handoff	Core connectivity	Management VLAN for infrastructure

The office printer sits on the office VLAN because it holds confidential documents. The receipt printer sits on the POS VLAN because it works alongside the POS terminal.

Design Decisions

Decision	Rationale
Four VLANs instead of one flat network	Smaller broadcast domains and clear security boundaries
Separate POS VLAN	Payment traffic is isolated from staff and guest devices
Separate guest VLAN plus ACL	Customers get internet only, never internal access
VLAN 99 for device management	Management traffic is kept apart from user traffic. Note that the Management Office VLAN (employees) and the Network Management VLAN (infrastructure) are different things
Router-on-a-stick	One physical link carries all VLANs, which is cost-effective for a small site
Trunk with allowed-VLAN list	Only required VLANs cross the link
Router-provided DHCP	Simple and appropriate for a site this size
SSH only	Telnet sends credentials in clear text
Pre-assigned spare ports	Adding devices later needs no reconfiguration

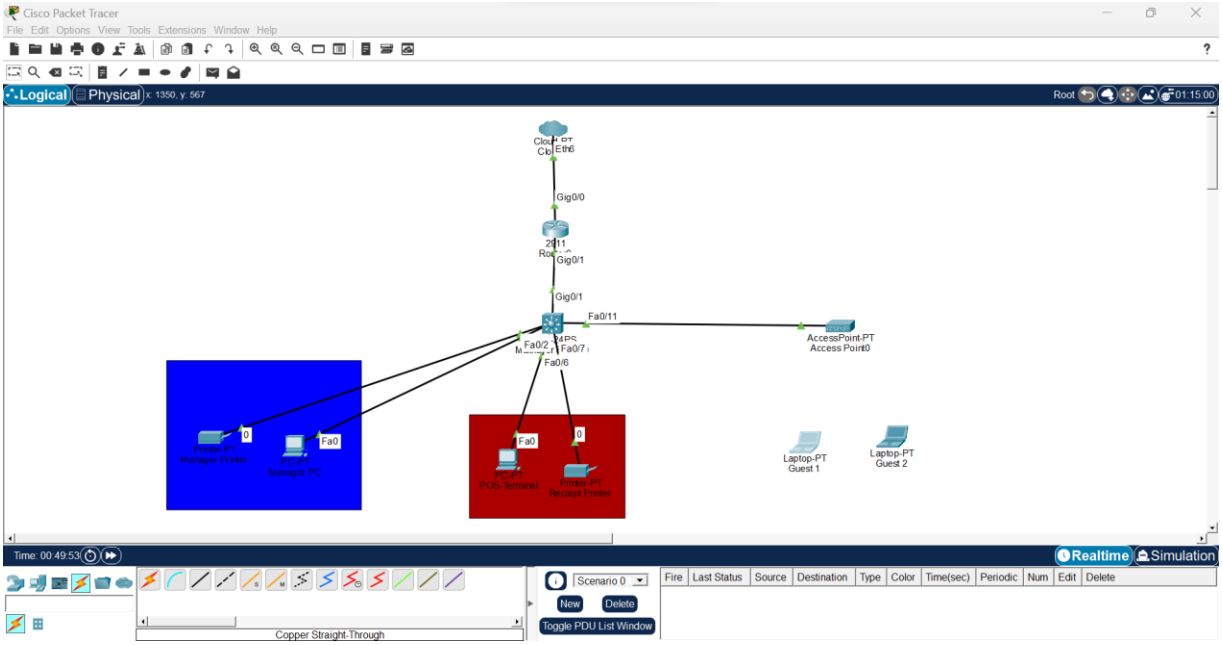
Packet Tracer Device List

Device	Model (suggested)
Router	Cisco 2911
Switch	Cisco 2960
Access point	Access Point-PT
Manager PC, POS terminal	PC
Office printer, receipt printer	Printer
Guest laptop	Laptop with wireless adapter module
ISP	Cloud or generic WAN device

Cabling

From	To	Cable
ISP	Router Gi0/0	Copper straight-through
Router Gi0/1	Switch Gi0/1	Copper straight-through (trunk)
Manager PC	Switch Fa0/1	Copper straight-through
Office printer	Switch Fa0/2	Copper straight-through
POS terminal	Switch Fa0/6	Copper straight-through
Receipt printer	Switch Fa0/7	Copper straight-through
Access point	Switch Fa0/11	Copper straight-through (PoE in real deployments)

In a real building these runs terminate at wall jacks and a patch panel in the network cabinet rather than plugging straight into the switch.



02 - Addressing and Port Map

Reference plan. The VLAN IDs and design come from the original walkthrough. Subnets, port numbers, DNS server and SSID below are a documented reference plan. If your Packet Tracer file differs, update this page. Documentation should always match the deployed network.

VLAN Plan

VLAN ID	Name	Purpose
10	MGMT-OFFICE	Manager PC and office printer
20	POS	POS terminal and receipt printer
30	GUEST-WIFI	Customer wireless
99	NETWORK-MGMT	Infrastructure management

IP Addressing

VLAN	Subnet	Gateway (router subinterface)	Reserved (excluded from DHCP)	DHCP range
10	192.168.10.0/24	192.168.10.1	.1 – .20	.21 – .254
20	192.168.20.0/24	192.168.20.1	.1 – .20	.21 – .254
30	192.168.30.0/24	192.168.30.1	.1 – .20	.21 – .254
99	192.168.99.0/24	192.168.99.1	Entire VLAN static	None

Reserving the first 20 addresses keeps room for gateways, printers and any future device that needs a static address.

Infrastructure Addresses

Device	Interface	Address
Router	Gi0/1.10	192.168.10.1
Router	Gi0/1.20	192.168.20.1
Router	Gi0/1.30	192.168.30.1
Router	Gi0/1.99	192.168.99.1
Switch	VLAN 99 SVI	192.168.99.2
Office printer	NIC	192.168.10.10 (static)
Receipt printer	NIC	192.168.20.10 (static)

Switch Port Map

Every port is documented. If the office printer goes offline in six months, nobody has to guess where it is plugged in.

Port	Device	Mode	VLAN	Notes
Fa0/1	Manager PC	Access	10	PortFast
Fa0/2	Office printer	Access	10	PortFast
Fa0/3 – 5	Spare (office)	Access	10	Pre-assigned for growth
Fa0/6	POS terminal	Access	20	PortFast
Fa0/7	Receipt printer	Access	20	PortFast
Fa0/8 – 10	Spare (POS)	Access	20	Pre-assigned for growth
Fa0/11	Wireless access point	Access	30	Lab simplification, see limitations
Gi0/1	Router Gi0/1	Trunk	10,20,30,99 allowed	Router-on-a-stick

Router Interface Map

Interface	Connected To	Purpose
Gi0/0	ISP	WAN (IP assignment deferred to a later lab)
Gi0/1	Switch Gi0/1	802.1Q trunk, no IP on the physical interface
Gi0/1.10 / .20 / .30 / .99	Logical	VLAN gateways

03- Configuration Walkthrough

The complete command sets live in ../configs/. This page explains the reasoning behind each stage, in the order the network was built.

Build Order

Physical installation and cabling

Switch: hardening, VLANs, ports, trunk, management, SSH

Router: hardening, WAN, subinterfaces, DHCP, ACL

End devices and wireless

Verification

Phase 1 — Physical Installation

Connect ISP to router, router to switch, then each end device to its documented port (see 02).

Consistent, documented port assignment makes later troubleshooting far easier.

Phase 2 — Switch

2.1 Baseline hardening

Command	Purpose
hostname CoffeeShop-SW1	Instantly identifies the device
no ip domain-lookup	Stops the CLI hanging when a command is mistyped and treated as a hostname
enable secret	Protects privileged EXEC mode with a hashed password
service password-encryption	Prevents plain-text passwords in the running config
banner motd	Legal warning against unauthorized access
Console password + login	Requires authentication even with physical console access

2.2 VLANs

Create VLANs 10, 20, 30 and 99 with descriptive names. Keep the distinction clear: VLAN 10 is where employees work, VLAN 99 is for managing network devices.

2.3 Port assignment

Use interface range to assign ports in blocks, including spare ports for future devices. Enable spanning-tree portfast only on ports connected to end devices. Never enable it on links between switches, as that can contribute to switching loops.

2.4 Trunk to router

interface GigabitEthernet0/1 switchport mode trunk switchport trunk allowed vlan 10,20,30,99

A trunk carries multiple VLANs over a single cable using 802.1Q tagging. Restricting the allowed list stops unnecessary VLAN traffic crossing the link.

2.5 Management interface

Assign an IP address to the VLAN 99 SVI and set ip default-gateway. A Layer 2 switch does not route, but it still needs a gateway to reply to management traffic from other subnets.

2.6 SSH

Set a domain name (required before generating keys)

Generate RSA keys (2048-bit)

Create a local admin account

Enable SSH version 2

Set VTY lines to transport input ssh with login local

Telnet is avoided because it sends usernames and passwords in clear text.

Phase 3 — Router

3.1 Baseline hardening

Same pattern as the switch: hostname, DNS lookup off, enable secret, password encryption, banner, console protection.

3.2 WAN interface

The interface is described and brought up, but no IP is assigned. Internet connectivity, NAT and the default route are intentionally out of scope for this lab.

3.3 Router-on-a-stick

The physical interface has no IP. Each VLAN gets a subinterface that acts as that VLAN's default gateway:

interface GigabitEthernet0/1.10 encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

encapsulation dot1Q <vlan> tells the router which VLAN tag the subinterface serves. Repeat for VLANs 20, 30 and 99.

3.4 DHCP

Exclude the first 20 addresses in each subnet, then create one pool per VLAN. Each pool defines the network, default router and DNS server.

For larger environments, DHCP is often provided by dedicated servers and forwarded with DHCP relay (ip helper-address). A router-hosted pool is perfectly adequate here.

3.5 Guest ACL

Without filtering, the router would happily route guest traffic into the office, POS and management networks. The GUEST-ACL fixes that:

ip access-list extended GUEST-ACL permit udp any eq bootpc any eq bootps deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255 deny ip 192.168.30.0 0.0.0.255 192.168.99.0 0.0.0.255 permit ip any any

Applied inbound on Gi0/1.30, so every packet from guest devices is inspected before the routing decision.

Rule order matters. The DHCP permit must come first. Guests' DHCP discover broadcasts would otherwise be dropped, leaving laptops with 169.254.x.x addresses.

Phase 4 — End Devices

Printers get static addresses from the reserved range so they are always in the same place.

The POS terminal and manager PC use DHCP.

The access point gets a descriptive SSID and a WPA2 passphrase.

The guest laptop needs a wireless adapter module installed before it can see Wi-Fi.

Details: ../configs/end-devices.md.

Phase 5 — Verification

Review both running configs for typos, then test connectivity. Never assume a configuration works just because no error appeared. See 04.

04 - Verification and Testing

Configuration is not finished until it has been tested against the design.

Configuration Review Commands

Device	Command	What to confirm
Switch	show vlan brief	VLANs 10/20/30/99 exist with correct ports
Switch	show interfaces trunk	Gi0/1 trunking, allowed VLANs 10,20,30,99
Switch	show ip interface brief	VLAN 99 SVI up with correct IP
Switch	show ip ssh	SSH enabled, version 2
Router	show ip interface brief	All four subinterfaces up/up
Router	show ip dhcp binding	Leases issued per VLAN
Router	show access-lists	GUEST-ACL rules in the right order, with hit counters
Both	show running-config	Check hostnames, descriptions and names for typos

If subinterfaces are down, check the trunk between router and switch first.

Connectivity Test Plan

Management office

#	Source	Destination	Expected
1	Manager PC	192.168.10.1 (gateway)	Success
2	Manager PC	Office printer (192.168.10.10)	Success
3	Manager PC	POS device (VLAN 20)	Success (inter-VLAN routing)

Point of sale

#	Source	Destination	Expected
4	POS terminal	192.168.20.1 (gateway)	Success
5	POS terminal	Receipt printer (192.168.20.10)	Success

Guest Wi-Fi

#	Source	Destination	Expected
6	Guest laptop	192.168.30.1 (gateway)	Success
7	Guest laptop	Office VLAN device	Blocked
8	Guest laptop	POS VLAN device	Blocked

Tests 7 and 8 prove the ACL works: guests reach their own network while internal resources stay protected.

Wireless DHCP Check

On the guest laptop run ipconfig. A 192.168.30.x address confirms DHCP and the ACL's DHCP permit are working. A 169.254.x.x address means the lease failed. See 05.

Not Tested

Internet reachability from the guest network is untested because the WAN side is not yet configured.

05 - Troubleshooting Guide

Symptom	Likely Cause	Fix
Guest laptop shows 169.254.x.x	DHCP permit missing, or placed after the deny rules in GUEST-ACL	Put permit udp any eq bootpc any eq bootps first, then run ipconfig /renew
Router subinterfaces are down	Trunk misconfigured or physical link down	Verify switchport mode trunk and the allowed VLAN list; check no shutdown on the router Gi0/1
A VLAN has no connectivity across the trunk	VLAN missing from the allowed list, or VLAN not created	show interfaces trunk, show vlan brief
Device gets no DHCP address (non-guest)	Wrong VLAN on port, pool misconfigured, or excluded range covers whole pool	show vlan brief, show ip dhcp pool
Device on the wrong subnet	Port assigned to the wrong VLAN	Check the port map and show vlan brief
SSH connection refused	Missing domain name, RSA keys, local user, or VTY not set to SSH	show ip ssh; confirm login local and transport input ssh
Cannot manage switch from another subnet	No default gateway on the switch	ip default-gateway 192.168.99.1
CLI hangs after a typo	DNS lookup enabled	no ip domain-lookup
Laptop cannot see Wi-Fi	No wireless adapter installed	Replace the Ethernet module with a wireless module
Printer unreachable	Wrong static IP or gateway	Compare against configs/end-devices.md

General Method

Confirm physical links and port status (show ip interface brief).

Confirm VLAN membership and trunking.

Confirm IP addressing and gateway.

Check DHCP leases and ACL hit counters (show access-lists).

Test from the closest point outward: gateway first, then other devices, then other VLANs.

06 - Limitations and Next Steps

Known Limitations

Limitation	Detail
No internet connectivity	The ISP-facing interface has no IP address, NAT or default route. The guest ACL permits internet traffic, but it cannot be tested yet
AP in the guest VLAN	The access point is both managed and used in VLAN 30. Production designs typically manage the AP from a dedicated management VLAN while clients are mapped to VLANs by SSID
Single router and switch	No redundancy
Router-on-a-stick throughput	One physical link carries all inter-VLAN traffic. Fine for a coffee shop, a bottleneck at scale
Router-based DHCP	Appropriate here, not for large environments
Simplified wireless	One SSID with a PSK. Enterprise wireless uses controllers, multiple APs and several SSIDs

Hardening Ideas

Shut down unused switch ports and place them in an unused "parking" VLAN

Change the native VLAN on the trunk to an unused ID

Add DHCP snooping and port security on access ports

Restrict VTY access to the management subnet with an access class

Add AAA with a central authentication server

Enable logging to a syslog server

Add an ACL or firewall policy for the POS VLAN, allowing only required payment destinations